

## **E-Commerce Website features**

### **1. User Authentication**

**Screens:** Registration Form, Login Page, Forgot Password

**Interaction:**

- Secure sign-up and login with encryption
- Email verification on registration
- Password reset via email link
- CAPTCHA to block automated sign-ups

### **2. Profile & Address Management**

**Screens:** Profile Overview, Edit Profile, Address Book

**Interaction:**

- View and update personal info and profile photo
- Change password securely
- Manage multiple delivery addresses

### **3. Role-Based Access Control**

**Screens:** Role Selector, Admin Panel, Seller Dashboard

**Interaction:**

- Assign roles: Admin, Buyer, Optional Seller
- Role-based navigation and permissions
- Sellers can manage their own product listings

### **4. Product Catalog with Variants**

**Screens:** Product Page, Variant Selector

**Interaction:**

- Products with size/color options
- Images and pricing change by variant
- Dynamic UI for variant selection

### **5. Advanced Search & Filtering**

**Screens:** Search Results, Filter Panel

**Interaction:**

- Search with real-time suggestions
- Filter by category, price, brand, etc.
- Instant update of search results

### **6. Cart & Save for Later**

**Screens:** Shopping Cart, Save for Later

**Interaction:**

- Add products with quantity controls
- Move items between cart and saved list
- Auto-update totals on cart changes

### **7. Wishlist Management**

**Screens:** Wishlist, Product Page Shortcut

**Interaction:**

- Mark favorites for future purchase

- Move items from wishlist to cart
- Wishlist sale alerts via email/app

## **8. Ratings & Reviews**

**Screens:** Product Reviews Section, Review Form

**Interaction:**

- Buyers submit ratings and comments
- Admin can moderate reviews
- Average rating shown on product

## **9. Flash Sales & Timed Deals**

**Screens:** Flash Sale Banner, Countdown Widget

**Interaction:**

- Limited-time deals with countdown timer
- Timer resets post-expiry
- Boosts urgency and conversions

## **10. Promo & Discount Codes**

**Screens:** Checkout Page, Admin Promo Panel

**Interaction:**

- Apply fixed or % promo codes
- Codes with expiry dates and usage limits
- Admin can manage and monitor codes

## **11. Multi-Payment Integration**

**Screens:** Checkout, Payment Confirmation

**Interaction:**

- Pay via cards, bKash, Nagad, or COD
- Secure encryption at checkout
- Instant on-screen and email confirmation

## **12. Order Tracking**

**Screens:** Order Status Tracker, Order Details

**Interaction:**

- Track: Confirmed → Shipped → Delivered
- Real-time status updates
- Shows estimated delivery window

## **13. Invoice & Billing**

**Screens:** Invoice Page, Download Section

**Interaction:**

- Auto-generated PDF invoices
- View/download from user dashboard
- Includes full breakdown of orders

## **14. Admin Dashboard**

**Screens:** Sales Overview, Product Manager, Reports Panel

**Interaction:**

- Manage users, orders, products, and coupons
- View stats and trends
- Export data for analysis

## **15. Notifications (Email & In-App)**

**Screens:** Notifications Center, Email Templates

**Interaction:**

- In-app alerts for status updates
- Emails for order events and promotions
- Password change notifications

## **16. Smart Product Recommendations**

**Screens:** Product Detail Page, Home Recommendations

**Interaction:**

- Suggest similar or related items
- Based on browsing/purchase behavior
- Enhances average order value

## **17. Customer Support & Contact**

**Screens:** Contact Form, Admin Inbox

**Interaction:**

- Submit queries via support form
- Admin receives and replies via email
- Simple 4-field form (name, email, subject, message)

## **18. Chatbot Assistant**

**Screens:** Chat Widget

**Interaction:**

- 24/7 rule-based assistant
- Answer FAQs, locate products, check order status
- Reduces load on support team

## **19. Loyalty & Reward System**

**Screens:** Rewards Dashboard, Checkout Points Panel

**Interaction:**

- Earn points with each order
- Redeem points for discounts
- View balance and redemption history

## **20. Order History & Invoicing**

**Screens:** Order History Page, Invoice Download

**Interaction:**

- Browse all previous orders
- Check status and download PDFs
- Includes price breakdown and shipping info

## **21. Returns & Refund Requests**

**Screens:** Return Request Form, Request Status Page

**Interaction:**

- Submit return/refund request
- Admin reviews and responds
- Status updates visible to users

## **22. Gift Cards & Vouchers**

**Screens:** Gift Card Purchase, Recipient Input

**Interaction:**

- Buy/send digital gift cards
- Recipient gets redeemable code
- Admin controls value and expiration

### **23. Activity Logs (Admin Only)**

**Screens:** Admin Logs Viewer

**Interaction:**

- Track user/admin activity logs
- View login times, edits, purchases
- Supports issue resolution and auditing

### **24. Data Export (Admin Only)**

**Screens:** Export Panel

**Interaction:**

- Export user/product/order data
- Choose CSV or PDF format
- Useful for reporting and backups

## User Types

- **User**
- **Guest User**
- **Admin**
- **Delivery Man**

Short Description of user Types

| Role         | Description & Permissions                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Guest User   | Browse products, search/filter, view reviews, add to cart or wishlist (local), but cannot place orders.                                         |
| User (Buyer) | Full user: login/register, manage profile, place orders, use wishlist/cart, earn rewards, track orders, write reviews, request returns/refunds. |
| Admin        | Manage everything: users, orders, products, promotions, analytics, access logs, resolve return/refund, export data, reply to support.           |
| Delivery Man | View assigned deliveries, update order status (e.g., “Out for delivery” → “Delivered”), see delivery address and contact.                       |

## Entity Overview

| Type           | Entity List                                           |
|----------------|-------------------------------------------------------|
| Core           | User, Role, Profile, Address, AuthToken               |
| Product        | Product, Variant, Image, Category, Brand, Review      |
| Order          | Cart, Order, Invoice, Return, Refund, DeliveryStatus  |
| Promotion      | Coupon, GiftCard, FlashSale, PromoCodeUsage           |
| Payment        | Payment, Transaction, PaymentMethod                   |
| Support        | SupportTicket, ChatMessage, Notification, ActivityLog |
| Recommendation | Recommendation, BrowsingHistory, SearchHistory        |
| Rewards        | RewardPoint, RewardTransaction                        |
| Optional       | DeliveryPerson, DeliveryAssignment                    |

## Details of each entity

### 1. User

| Column Name   | Type             | Description               |
|---------------|------------------|---------------------------|
| id (PK)       | UUID/INT         | Unique user ID            |
| name          | VARCHAR          | Full name                 |
| email         | VARCHAR (unique) | Login email               |
| phone         | VARCHAR          | Phone number              |
| password_hash | TEXT             | Encrypted password        |
| is_verified   | BOOLEAN          | Email verification status |
| created_at    | TIMESTAMP        | Registration date         |

### 2. Role

| Column Name | Type    | Description                     |
|-------------|---------|---------------------------------|
| id (PK)     | INT     | Role ID                         |
| name        | VARCHAR | Role name: Admin, Buyer, Seller |

### 3. UserRole

| Column Name  | Type     | Description         |
|--------------|----------|---------------------|
| id (PK)      | INT      | ID                  |
| user_id (FK) | UUID/INT | References User(id) |
| role_id (FK) | INT      | References Role(id) |

#### 4. Profile

| Column Name   | Type     | Description         |
|---------------|----------|---------------------|
| id (PK)       | UUID/INT | Profile ID          |
| user_id (FK)  | UUID/INT | References User(id) |
| gender        | VARCHAR  | Gender              |
| dob           | DATE     | Date of Birth       |
| profile_photo | TEXT     | Image URL or path   |
| bio           | TEXT     | Short description   |

#### 5. Address

| Column Name    | Type     | Description               |
|----------------|----------|---------------------------|
| id (PK)        | INT      | Address ID                |
| user_id (FK)   | UUID/INT | References User(id)       |
| recipient_name | VARCHAR  | Name for delivery         |
| phone          | VARCHAR  | Contact phone             |
| address_line   | TEXT     | Street or address details |
| city           | VARCHAR  | City name                 |
| area           | VARCHAR  | Local area                |
| postal_code    | VARCHAR  | ZIP or postal code        |
| is_default     | BOOLEAN  | Default address flag      |

#### 6. Product

| Column Name      | Type      | Description                  |
|------------------|-----------|------------------------------|
| id (PK)          | INT       | Product ID                   |
| title            | VARCHAR   | Product name                 |
| description      | TEXT      | Detailed description         |
| base_price       | DECIMAL   | Starting price               |
| brand_id (FK)    | INT       | References Brand(id)         |
| category_id (FK) | INT       | References Category(id)      |
| created_by (FK)  | UUID/INT  | References User(id) (seller) |
| created_at       | TIMESTAMP | Date added                   |

#### 7. ProductVariant

| Column Name     | Type    | Description              |
|-----------------|---------|--------------------------|
| id (PK)         | INT     | Variant ID               |
| product_id (FK) | INT     | References Product(id)   |
| variant_name    | VARCHAR | e.g., Red - M, Blue - XL |
| price           | DECIMAL | Variant price            |
| stock           | INT     | Available quantity       |
| sku             | VARCHAR | Stock Keeping Unit       |

## 8. ProductImage

| Column Name     | Type           | Description                              |
|-----------------|----------------|------------------------------------------|
| id (PK)         | INT            | Image ID                                 |
| product_id (FK) | INT            | References Product(id)                   |
| variant_id (FK) | INT (nullable) | Optionally references ProductVariant(id) |
| image_url       | TEXT           | Image file path or URL                   |

## 9. Category

| Column Name    | Type           | Description                    |
|----------------|----------------|--------------------------------|
| id (PK)        | INT            | Category ID                    |
| name           | VARCHAR        | Category name                  |
| parent_id (FK) | INT (nullable) | Self-reference for subcategory |

## 10. Brand

| Column Name | Type    | Description            |
|-------------|---------|------------------------|
| id (PK)     | INT     | Brand ID               |
| name        | VARCHAR | Brand name             |
| logo_url    | TEXT    | Logo image             |
| description | TEXT    | Optional brand details |

## 11. Review

| Column Name     | Type      | Description             |
|-----------------|-----------|-------------------------|
| id (PK)         | INT       | Review ID               |
| user_id (FK)    | UUID/INT  | Reviewer                |
| product_id (FK) | INT       | Reviewed product        |
| rating          | INT (1-5) | Star rating             |
| comment         | TEXT      | Review text             |
| created_at      | TIMESTAMP | Review time             |
| approved        | BOOLEAN   | Admin moderation status |

## 12. UserCart

| Column Name     | Type      | Description                   |
|-----------------|-----------|-------------------------------|
| id (PK)         | INT       | Unique cart item ID           |
| user_id (FK)    | UUID/INT  | References User(id)           |
| variant_id (FK) | INT       | References ProductVariant(id) |
| quantity        | INT       | Selected quantity             |
| created_at      | TIMESTAMP | When item was added           |
| updated_at      | TIMESTAMP | Last modification time        |

## 13. Wishlist

| Column Name  | Type     | Description         |
|--------------|----------|---------------------|
| id (PK)      | INT      | Wishlist ID         |
| user_id (FK) | UUID/INT | References User(id) |

|                 |           |                        |
|-----------------|-----------|------------------------|
| product_id (FK) | INT       | References Product(id) |
| created_at      | TIMESTAMP | Added date             |

#### 14. Order

| Column Name     | Type      | Description                                 |
|-----------------|-----------|---------------------------------------------|
| id (PK)         | INT       | Order ID                                    |
| user_id (FK)    | UUID/INT  | Buyer                                       |
| address_id (FK) | INT       | Delivery address                            |
| total_amount    | DECIMAL   | Total cost                                  |
| status          | VARCHAR   | e.g., Confirmed, Shipped                    |
| created_at      | TIMESTAMP | Order date                                  |
| current_status  | VARCHAR   | Latest status (Confirmed/Shipped/Delivered) |
| status_history  | JSONB     | Array of {status, timestamp, updated_by}    |

*Example status\_history value:*

```
[
  {"status": "Packed", "timestamp": "2023-10-01T10:00:00Z", "updated_by":
"admin@email.com"},
  {"status": "Shipped", "timestamp": "2023-10-02T09:30:00Z", "updated_by": "system"}
]
```

#### 15. OrderItem

| Column Name     | Type    | Description                   |
|-----------------|---------|-------------------------------|
| id (PK)         | INT     | Order item ID                 |
| order_id (FK)   | INT     | References Order(id)          |
| variant_id (FK) | INT     | References ProductVariant(id) |
| quantity        | INT     | Quantity ordered              |
| unit_price      | DECIMAL | Price per unit                |

#### 16. Invoice

| Column Name   | Type      | Description           |
|---------------|-----------|-----------------------|
| id (PK)       | INT       | Invoice ID            |
| order_id (FK) | INT       | References Order(id)  |
| pdf_url       | TEXT      | Invoice file location |
| generated_at  | TIMESTAMP | Date of generation    |

#### 17. ReturnRequest

| Column Name        | Type      | Description               |
|--------------------|-----------|---------------------------|
| id (PK)            | INT       | Return ID                 |
| order_item_id (FK) | INT       | References OrderItem(id)  |
| reason             | TEXT      | Return reason             |
| status             | VARCHAR   | Pending/Approved/Rejected |
| created_at         | TIMESTAMP | Submission time           |

#### 18. Refund

| Column Name | Type | Description |
|-------------|------|-------------|
| id (PK)     | INT  | Refund ID   |

|                |         |                              |
|----------------|---------|------------------------------|
| return_id (FK) | INT     | References ReturnRequest(id) |
| amount         | DECIMAL | Refunded amount              |
| method         | VARCHAR | bKash/Card/etc.              |
| status         | VARCHAR | e.g., Processing, Done       |

## 19. Coupon

| Column Name | Type    | Description       |
|-------------|---------|-------------------|
| id (PK)     | INT     | Coupon ID         |
| code        | VARCHAR | Unique code       |
| type        | VARCHAR | Flat / Percentage |
| value       | DECIMAL | Discount value    |
| expiry_date | DATE    | Expiration date   |
| usage_limit | INT     | Max uses          |

## 20. GiftCard

| Column Name     | Type    | Description       |
|-----------------|---------|-------------------|
| id (PK)         | INT     | Gift card ID      |
| code            | VARCHAR | Redeemable code   |
| amount          | DECIMAL | Gift card balance |
| recipient_email | VARCHAR | Receiver's email  |
| sender_name     | VARCHAR | Sender name       |
| expiry_date     | DATE    | Expiration date   |

## 21. Payment

| Column Name    | Type    | Description             |
|----------------|---------|-------------------------|
| id (PK)        | INT     | Payment ID              |
| order_id (FK)  | INT     | References Order(id)    |
| method         | VARCHAR | bKash, Nagad, Card, COD |
| amount         | DECIMAL | Amount paid             |
| status         | VARCHAR | Paid / Failed / Pending |
| transaction_id | VARCHAR | External gateway ID     |

## 22. Notification

| Column Name  | Type      | Description              |
|--------------|-----------|--------------------------|
| id (PK)      | INT       | Notification ID          |
| user_id (FK) | UUID/INT  | Recipient                |
| type         | VARCHAR   | e.g., OrderUpdate, Promo |
| message      | TEXT      | Notification content     |
| read         | BOOLEAN   | Read/unread              |
| created_at   | TIMESTAMP | Timestamp                |

## 23. AuthToken

| Column Name  | Type     | Description              |
|--------------|----------|--------------------------|
| id (PK)      | INT      | Unique token ID          |
| user_id (FK) | INT      | References User(id)      |
| token        | TEXT     | Authentication token     |
| created_at   | DATETIME | Token creation time      |
| expires_at   | DATETIME | Expiry time of the token |

#### 24. DeliveryPerson

| Column Name | Type    | Description                 |
|-------------|---------|-----------------------------|
| id (PK)     | INT     | Unique delivery person ID   |
| name        | VARCHAR | Full name                   |
| phone       | VARCHAR | Contact number              |
| email       | VARCHAR | Email address               |
| address     | TEXT    | Residential or base address |
| is_active   | BOOLEAN | Availability status         |

#### 25. DeliveryAssignment

| Column Name             | Type     | Description                        |
|-------------------------|----------|------------------------------------|
| id (PK)                 | INT      | Unique assignment ID               |
| order_id (FK)           | INT      | References Order(id)               |
| delivery_person_id (FK) | INT      | References DeliveryPerson(id)      |
| assigned_at             | DATETIME | Assignment timestamp               |
| status                  | VARCHAR  | Assignment status (e.g., Assigned) |

#### 26. FlashSale

| Column Name     | Type     | Description                         |
|-----------------|----------|-------------------------------------|
| id (PK)         | INT      | Unique flash sale ID                |
| product_id (FK) | INT      | References Product(id)              |
| discount        | DECIMAL  | Discount value (flat or percentage) |
| start_time      | DATETIME | Sale start time                     |
| end_time        | DATETIME | Sale end time                       |
| is_active       | BOOLEAN  | Status of flash sale                |

#### 27. PromoCodeUsage

| Column Name    | Type     | Description                   |
|----------------|----------|-------------------------------|
| id (PK)        | INT      | Unique promo usage ID         |
| user_id (FK)   | INT      | References User(id)           |
| coupon_id (FK) | INT      | References Coupon(id)         |
| used_at        | DATETIME | Timestamp when promo was used |

#### 28. Transaction

| Column Name     | Type     | Description              |
|-----------------|----------|--------------------------|
| id (PK)         | INT      | Unique transaction ID    |
| payment_id (FK) | INT      | References Payment(id)   |
| request_data    | TEXT     | Raw data sent to gateway |
| response_data   | TEXT     | Response received        |
| timestamp       | DATETIME | Time of transaction      |

### 29. PaymentMethod

| Column Name | Type    | Description                 |
|-------------|---------|-----------------------------|
| id (PK)     | INT     | Unique method ID            |
| name        | VARCHAR | e.g., bKash, Card, COD      |
| description | TEXT    | Optional details            |
| is_active   | BOOLEAN | Whether method is available |

### 30. SupportTicket

| Column Name  | Type     | Description                     |
|--------------|----------|---------------------------------|
| id (PK)      | INT      | Ticket ID                       |
| user_id (FK) | INT      | References User(id)             |
| subject      | VARCHAR  | Ticket subject                  |
| message      | TEXT     | Initial user message            |
| status       | VARCHAR  | e.g., Open, In Progress, Closed |
| created_at   | DATETIME | Ticket creation time            |
| resolved_at  | DATETIME | When it was resolved (nullable) |

### 31. ChatMessage

| Column Name            | Type     | Description                            |
|------------------------|----------|----------------------------------------|
| id (PK)                | INT      | Message ID                             |
| user_id (FK)           | INT      | References User(id)                    |
| sender_type            | VARCHAR  | 'user', 'admin', or 'bot'              |
| message                | TEXT     | Chat content                           |
| timestamp              | DATETIME | Message timestamp                      |
| support_ticket_id (FK) | INT      | Optional, references SupportTicket(id) |

### 32. ActivityLog

| Column Name  | Type     | Description                        |
|--------------|----------|------------------------------------|
| id (PK)      | INT      | Log entry ID                       |
| user_id (FK) | INT      | References User(id)                |
| action       | VARCHAR  | e.g., Login, Logout, Edit Product  |
| metadata     | TEXT     | Additional data (IP, device, etc.) |
| timestamp    | DATETIME | When action happened               |

### 33. Recommendation

| Column Name     | Type    | Description                       |
|-----------------|---------|-----------------------------------|
| id (PK)         | INT     | Unique recommendation ID          |
| user_id (FK)    | INT     | References User(id)               |
| product_id (FK) | INT     | Suggested Product                 |
| reason          | TEXT    | e.g., "Based on browsing history" |
| score           | DECIMAL | Confidence/priority score         |

### 34. BrowsingHistory

| Column Name     | Type     | Description         |
|-----------------|----------|---------------------|
| id (PK)         | INT      | Unique record ID    |
| user_id (FK)    | INT      | References User(id) |
| product_id (FK) | INT      | Product viewed      |
| viewed_at       | DATETIME | Timestamp of view   |

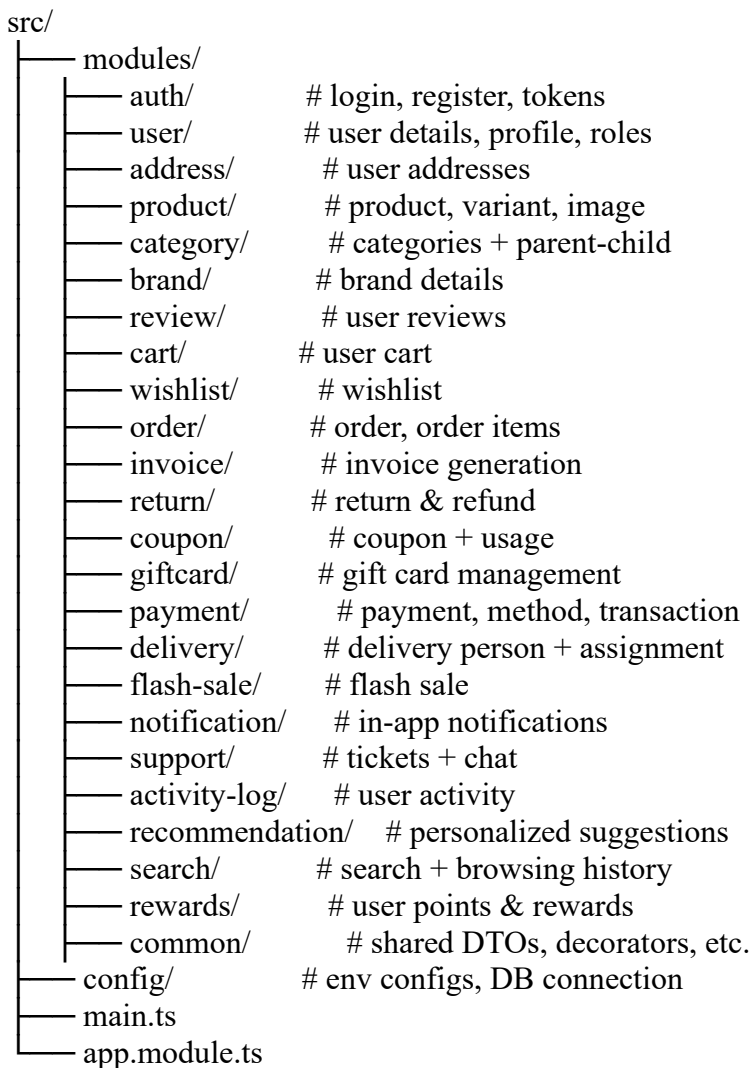
### 35. SearchHistory

| Column Name  | Type     | Description          |
|--------------|----------|----------------------|
| id (PK)      | INT      | Unique record ID     |
| user_id (FK) | INT      | References User(id)  |
| query        | VARCHAR  | Search query entered |
| searched_at  | DATETIME | When search was made |

### 36. UserRewards

| Column Name   | Type           | Description                        |
|---------------|----------------|------------------------------------|
| id (PK)       | INT            | Transaction ID                     |
| user_id (FK)  | UUID/INT       | References User(id)                |
| balance       | INT            | Current points after transaction   |
| points        | INT            | +/- points change                  |
| reason        | VARCHAR        | "Purchase"/"Redemption"/"Expiry"   |
| order_id (FK) | INT (nullable) | References Order(id) if applicable |
| created_at    | TIMESTAMP      | Transaction time                   |

## Recommended Folder Structure



### API Endpoint Structure

Each module should follow RESTful conventions. Examples:

#### ☒ Auth Module

POST /auth/register  
POST /auth/login  
POST /auth/logout  
POST /auth/verify-email  
POST /auth/refresh-token

#### User Module

GET /users/me  
PATCH /users/me  
GET /users/:id

#### Role Module

GET /roles  
POST /user-role

## Address Module

GET /addresses  
POST /addresses  
PATCH /addresses/:id  
DELETE /addresses/:id

## Product Module

GET /products  
GET /products/:id  
POST /products (Seller)  
PATCH /products/:id (Seller)  
DELETE /products/:id (Seller)

## Product Variant + Images

GET /products/:id/variants  
POST /products/:id/variants  
GET /variants/:id/images  
POST /variants/:id/images

## Category Module

GET /categories  
GET /categories/:id

## Brand Module

GET /brands

## Review Module

GET /products/:id/reviews  
POST /reviews  
PATCH /reviews/:id/approve (Admin)

## Cart Module

GET /cart  
POST /cart  
PATCH /cart/:id  
DELETE /cart/:id

## Wishlist Module

GET /wishlist  
POST /wishlist  
DELETE /wishlist/:id

## Order Module

GET /orders  
POST /orders  
GET /orders/:id

## Invoice Module

GET /orders/:id/invoice  
POST /return-requests  
PATCH /return-requests/:id  
POST /refunds

## Coupon + Promo Usage

GET /coupons  
POST /coupons/apply

## Gift Card

GET /giftcards

POST /giftcards/redeem

### **Payment Module**

bash

CopyEdit

POST /payments/initiate

POST /payments/callback

GET /payments/:id

### **Delivery**

GET /delivery-person

POST /delivery-assignment

### **Flash Sale**

GET /flash-sales

### **Notification**

GET /notifications

PATCH /notifications/:id/read

### **Support Ticket + Chat**

POST /support/tickets

GET /support/tickets/:id/messages

POST /support/tickets/:id/messages

### **Activity Logs**

GET /activity-logs

### **Recommendation**

GET /recommendations

### **Search + Browsing + History**

GET /search?query=...

GET /browsing-history

### **Rewards**

GET /rewards

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## **Suggestions for Efficient Team Workflow**

### **1. Monorepo with Nx or TurboRepo (optional)**

Helps manage large-scale codebase with isolated modules.

### **2. Swagger (OpenAPI) Integration**

- Auto-generate and document APIs using @nestjs/swagger.

### **3. Use DTOs & Validation Pipes**

- Centralize validation logic with class-validator in DTOs.

### **4. Role-Based Access Control (RBAC)**

- Use Guards to protect admin-only or seller-only endpoints.

### **5. Git Workflow**

- Branching strategy:
  - main: production
  - dev: staging
  - feature/xyz: development

### **6. Postman/Insomnia Shared Workspace**

- Let frontend/backend teams test endpoints collaboratively.

### **7. Jira/Notion + GitHub Projects for Task Tracking**

- Assign specific module ownership.
- Example:
  - Module: Product → Dev A

- Module: Order → Dev B

## 8. DB Versioning Tool

- Use **TypeORM migrations** or **Prisma Migrate** (if considered later).
- Automate schema updates.

## 9. Unit + E2E Testing

- Use Jest for unit tests.
- Use supertest with NestJS for e2e routes.

## 10. Code Lint + Prettier + Husky

- Ensure consistency with automated lint checks before commit.

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## Bonus Tips

- Use **Redis** for session, cache, and rate limiting.
- Integrate **BullMQ** or **RabbitMQ** for background tasks (email, refunds).
- Log activities with **Winston** or **Pino**.
- Track errors with **Sentry**.
- Protect rate abuse with **Nest Rate Limiter**.